

Initial Project Proposal

Pharmacy Management System

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1. Project Description

The Pharmacy Management System (PMS) is a web-based software application designed to automate the daily operational activities of a retail pharmacy. The system is intended to replace traditional paper-based and spreadsheet-based methods of managing medicines, stock levels, suppliers, purchases, sales, and customer information with a centralized, digital platform that is accessible to authorized staff members.

The system works by allowing pharmacy staff to record and manage information related to medicines, monitor stock quantities in real time, process purchase orders from suppliers, handle daily sales transactions with customers, and maintain accurate records of all these activities within a single database. Different categories of users, such as administrators, pharmacists, and pharmacy staff, are given specific levels of access according to their responsibilities, ensuring that the system is both secure and easy to use.

The primary purpose of the Pharmacy Management System is to reduce the reliance on manual record-keeping, minimize human error, and provide pharmacy management with quick and reliable access to information such as current stock levels, expiring medicines, sales history, and supplier details. By automating these routine tasks, the system is expected to improve the overall efficiency, accuracy, and reliability of daily pharmacy operations, allowing staff to focus more on customer service and less on administrative paperwork.

2. Problem Statement

In many small and medium-sized pharmacies, core business activities such as maintaining medicine records, tracking inventory, recording supplier purchases, processing sales transactions, and managing customer information are still carried out manually or using basic spreadsheet tools. While this approach may be manageable when the scale of operations is small, it becomes increasingly difficult to sustain as the volume of medicines, transactions, and records grows.

Manual management of medicine records often leads to inconsistent or outdated information, making it difficult for staff to quickly verify details such as medicine composition, dosage, pricing, or expiry dates. Without a centralized system, inventory levels are frequently inaccurate, which can result in stock shortages of essential medicines or, conversely, overstocking of items that are rarely used, both of which negatively affect the pharmacy's efficiency and profitability.

Manual purchase and supplier record-keeping can result in duplicated orders, missed deliveries, or difficulty in tracking which supplier provided a particular batch of medicine, especially when issues such as defective stock or delayed payments arise. Similarly, manually processed sales transactions are more prone to calculation errors, misplaced records, and delays in generating accurate daily or monthly sales figures.

In addition, manually maintained customer records make it difficult to track purchase history or provide personalized service, and generating reports for management decision-making, such as identifying best-selling medicines or slow-moving stock, becomes a time-consuming and error-prone process when performed manually. These challenges collectively highlight the need for a dedicated, computerized Pharmacy Management System that can centralize records, reduce manual effort, minimize errors, and provide accurate, real-time information to support efficient pharmacy operations and informed decision-making.

3. Project Objectives

Overall Aim

The overall aim of this project is to design and develop a web-based Pharmacy Management System that automates the core operational activities of a pharmacy, including medicine management, inventory control, supplier and purchase management, sales processing, customer record-keeping, and report generation, in order to improve accuracy, efficiency, and reliability in daily pharmacy operations.

Specific Objectives

- To develop a secure user login and access control mechanism that restricts system functions according to user roles.
- To design a centralized database for storing and managing medicine information accurately and consistently.
- To implement a real-time inventory and stock management module that reduces the risk of stock shortages and overstocking.
- To develop a supplier management module that maintains organized records of suppliers and their supplied products.
- To create a purchase management module that records and tracks purchase orders and incoming stock from suppliers.
- To develop a sales management module that enables accurate and efficient processing of customer transactions.
- To maintain customer records that support better service and traceability of purchase history.
- To provide a reporting module that generates useful summaries to support management decision-making.
- To reduce the amount of manual paperwork and human error involved in day-to-day pharmacy operations.

4. Target Users

The Pharmacy Management System is designed to be used by three main categories of users, each with distinct responsibilities and levels of access within the system.

Administrator / Manager

The Administrator or Manager holds the highest level of access within the system and is primarily responsible for overall system supervision. Their responsibilities include creating and managing user accounts for pharmacists and pharmacy staff, assigning appropriate access permissions, overseeing supplier and purchase records, monitoring overall stock and sales performance, and reviewing system-generated reports to make informed business decisions. The Administrator is also responsible for ensuring that the pharmacy's operations, as reflected in the system, comply with organizational policies.

Pharmacist

The Pharmacist is responsible for managing medicine-related information within the system, including verifying medicine details, dosage information, and expiry dates. The Pharmacist reviews and approves critical stock-related decisions, such as identifying medicines that need to be reordered or removed due to expiry, and may also oversee sales transactions to ensure that medicines are dispensed correctly and in compliance with pharmaceutical regulations.

Pharmacy Staff

Pharmacy Staff, such as sales assistants and store attendants, primarily use the system to perform daily operational tasks, including processing sales transactions, updating stock quantities as items are sold or received, recording customer purchases, and assisting with routine inventory checks. Their access within the system is generally limited to functions required for daily transactional activities, ensuring that sensitive administrative functions remain protected.

5. Main System Features

The Pharmacy Management System will consist of the following main modules, each designed to handle a specific area of pharmacy operations while remaining interconnected to ensure smooth and consistent data flow across the system.

User Login and User Management

This module provides secure authentication for all users of the system through a login mechanism that verifies usernames and passwords. It also enables the Administrator to create, update, deactivate, or remove user accounts and to assign role-based access permissions to Administrators, Pharmacists, and Pharmacy Staff. This module acts as the entry point to the system and controls which features each user category is permitted to access, thereby maintaining data security and accountability throughout the system.

Medicine Management

The Medicine Management module is responsible for maintaining detailed information about each medicine offered by the pharmacy, such as medicine name, category, dosage form, manufacturer, unit price, and expiry date. This module allows authorized users to add new medicines, update existing medicine details, and remove discontinued items from the catalogue. It serves as the master reference for medicine data and provides essential information to the Inventory, Purchase, and Sales modules, ensuring that all other modules work with consistent and up-to-date medicine details.

Inventory / Stock Management

The Inventory / Stock Management module tracks the actual quantity of each medicine available in the pharmacy at any given time. It updates stock levels automatically whenever a purchase is received or a sale is made, and it provides alerts for low-stock items or medicines nearing expiry. While the Medicine Management module stores descriptive details about each medicine, this module focuses specifically on quantity tracking and stock movement, working closely with the Purchase and Sales modules to ensure that stock records remain accurate and current.

Supplier Management

This module maintains records of the suppliers from whom the pharmacy purchases its medicines, including supplier name, contact details, and the list of medicines they typically supply. It allows authorized users to add, update, or remove supplier records. Accurate supplier information supports the Purchase Management module by making it easier to identify the appropriate supplier when placing new orders and to maintain accountability in case of stock quality issues or delivery concerns.

Purchase Management

The Purchase Management module handles the process of ordering medicines from suppliers and recording incoming stock. It allows staff to create purchase records that specify the supplier, medicines ordered, quantities, and purchase price. Once a purchase is confirmed as received, this module updates the Inventory / Stock Management module accordingly, ensuring that newly received stock is accurately reflected in the system and is immediately available for sale.

Sales Management

The Sales Management module manages the process of recording and processing customer transactions at the point of sale. It allows pharmacy staff to select medicines, specify quantities, and generate a sales invoice or receipt. Upon completion of a sale, this module automatically updates the Inventory / Stock Management module to reflect the reduced quantity of medicines and links the transaction to the Customer Management module when applicable, ensuring accurate and traceable sales records.

Customer Management

This module maintains basic records of customers who make purchases, such as customer name and contact information, along with a history of their past purchases where relevant. It supports the Sales Management module by allowing staff to associate transactions with specific customers, which can be useful for tracking purchase history and improving customer service.

Report Generation

The Report Generation module compiles data from the other modules to produce meaningful summaries for management review. Reports may include daily or monthly sales summaries, current stock levels, low-stock and expiry alerts, and purchase history by supplier. By drawing information from the Medicine, Inventory, Purchase, and Sales modules, this feature provides management with a consolidated view of pharmacy performance to support informed decision-making.

6. Proposed Technology Stack

The following technologies have been selected for the development of the Pharmacy Management System, based on their suitability for building a reliable, web-based data management application.

Frontend: HTML, CSS, and JavaScript will be used to design and implement the user interface. These technologies are well-suited for creating clear, responsive, and interactive forms and dashboards that pharmacy staff can use easily, without requiring specialized software on the client side.

Backend: PHP will be used to implement the server-side logic of the system, including handling user authentication, processing form submissions, and managing communication between the user interface and the database. PHP is widely supported, integrates smoothly with MySQL, and is appropriate for developing a data-driven web application of this scale.

Database: MySQL will be used to store and manage all system data, including medicine records, stock levels, supplier details, purchase and sales transactions, and customer information. MySQL is a reliable relational database management system that supports structured data storage and efficient querying, which is essential for maintaining accurate pharmacy records.

Development Environment: XAMPP will be used to provide a local server environment, including Apache and MySQL, for developing and testing the system before deployment. Visual Studio Code will be used as the primary code editor due to its lightweight design and strong support for PHP, HTML, CSS, and JavaScript development.

7. Expected Outcome

Upon completion, the Pharmacy Management System is expected to provide a fully functional, web-based solution that replaces manual record-keeping with a centralized and reliable digital platform. The system should enable pharmacy staff to manage medicine information, monitor stock levels, and process purchases and sales with greater speed and accuracy than manual methods.

The completed system is expected to improve the overall efficiency of daily pharmacy operations by reducing the time and effort required to record and retrieve information. It should also improve the accuracy of records by minimizing human error associated with manual data entry and calculations. In terms of stock monitoring, the system is expected to provide real-time visibility of medicine quantities, helping to prevent both stock shortages and unnecessary overstocking.

Additionally, the system is expected to strengthen record management by maintaining organized and easily accessible information related to medicines, suppliers, purchases, sales, and customers. The report generation feature should allow management to make more informed decisions based on accurate, up-to-date summaries of pharmacy performance. Overall, the Pharmacy Management System is expected to significantly reduce the reliance on manual, paper-based processes and contribute to more efficient, accurate, and reliable pharmacy operations.
